Soham Bagchi

PhD Student @ University of Utah | Open-Source Kernel Developer

sohambagchi@outlook.com | +1 (801) 649-9657 | sohambagchi.com | github.com/sohambagchi | linkedin.com/in/soham-bagchi

Education

PhD in Computer Science — Salt Lake City, Utah
The University of Utah — 2023–Present
GPA: 3.83 / 4.0.

B.Sc. in Computer Science — Sonipat, Haryana
Ashoka University — 2019–2023
UG GPA: 3.83 / 4.0; ASP GPA: 3.71 / 4.0.

ISC / ICSE — Gurgaon, Haryana
The Shri Ram School — 2019
ICSE: 95.0%; ISC: 90.25%.

Publications

Formalizing the User–Kernel Shared Memory Boundary — ASPLOS '26
Hastin Kapoor, **Soham Bagchi**, An Qi Zhang, Soham Chakraborty, Marco Elver, Andrés Goens, Susmit Sarkar, Ryan Stutsman, and Vijay Nagarajan

Beyond Zero-Cost: Understanding Rust Safety Overheads in Systems Code — PLOS '26
Soham Bagchi, Manvik Nanda, and Anton Burtsev

Consistency and Coherence of the NVIDIA Grace-Hopper Superchip — ISMM '26
Soham Bagchi, Sanya Srivastava, Reese Levine, Tyler Sorensen, Ryan Stutsman, and Vijay Nagarajan
doi.org/10.1145/3814942.3816134

Research & Experience

PhD Student — Salt Lake City, Utah
The University of Utah — Aug. 2023–Present
Prof. Anton Burtsev (2026–present): Agentic AI for developing isolated and performant userspace drivers. Prof. Vijay Nagarajan (2023–2026): Compound Memory Consistency Models for heterogeneous hardware and software. Areas: Computer Systems; Computer Architecture; LLMs; Agentic Systems; Network Drivers.

Cluster Administrator — HPC — Sonipat, Haryana
Ashoka University — Apr. 2022–Dec. 2022
Part of a team that serviced and maintained Ashoka University's high-performance computing cluster. Areas: HPC; Linux.

Research Assistant — Sonipat, Haryana
Ashoka University — Jun. 2021–Aug. 2022
With Prof. Manu Awasthi, analyzed file-metadata lookups to design optimizations for non-volatile memory. Areas: Storage Systems; Non-Volatile Memory.

Research Projects

High-Performance Userspace Drivers — 2026–Present
C; Rust; DPDK
Compared C and Rust userspace drivers for the 100 Gb E810 NIC to measure overheads introduced by Rust safety features.

C-LKMM Compound Memory Model — 2025–2026
C11; Linux Kernel; eBPF
Implemented high-performance eBPF data structures compliant with the C-LKMM Compound Memory Model; submitted bug-fix patches to liburing and kcov.

Heterogeneous Shared Memory — 2024–2025
CUDA; NVLink-C2C; GH200
Verified hardware memory-consistency models on heterogeneous shared-memory systems and reverse-engineered heterogeneous coherence protocols.

Distributed Orchestration — 2023
Kubernetes; CXL; etcd
Explored shared-memory control-plane algorithms for containerized and serverless workloads.

Teaching

Mini-Practicum — Undergraduate Computer Organization Spring 2026
Prof. Rajeev Balasubramonian
Taught two lectures on The Cache and Memory Hierarchy as part of Prof. Erin Parker's Teaching CS in Higher Education.

Graduate TA — Operating Systems Spring 2025
Prof. Anton Burtsev
Taught a lecture on Linking and Loading.

Graduate TA — Computer Architecture Fall 2024
Prof. Vijay Nagarajan
Taught a lecture on Instruction Set Architectures.

Undergraduate TA — Operating Systems Spring 2023
Prof. Manu Awasthi

Undergraduate TA — Introduction to Computer Programming Spring 2020
Prof. Goutam Paul

Leadership & Service

President — GradSAC 2024–25
Kahlert School of Computing

Secretary — EGSC 2024–25
Price College of Engineering

Vice-President — GradSAC 2023–24
Kahlert School of Computing

President — The Computer Science Society 2021–22
Ashoka University

Member — Women in Computing Society 2021–22
Ashoka University

Marketing Team Head — RedBrick Hacks 2020–22
Ashoka University

Awards

Kahlert Impact Prize, Honorable Mention Apr. 2026
Kahlert School of Computing
For contributions to the department and college as part of GradSAC and EGSC.

Teaching Assistant Excellence Award Jul. 2023
CS Department, Ashoka University
For Operating Systems S'23.

Best Capstone Project Award Jul. 2023
CS Department, Ashoka University
For HashPath: Analysis of Metadata Caches in the Presence of Hybrid Memory Hierarchies.

Research Excellence Award Jul. 2022
CS Department, Ashoka University

Teaching Assistant Excellence Award Jul. 2021
CS Department, Ashoka University
For Introduction to Computer Programming S'20.

Dean's List Semesters 2–6
Ashoka University
For maintaining a GPA of 3.65 or higher.

Additional Information

Technical Skills
Proficient: Python (multiprocessing, threading); C++; C; CUDA; Javascript; Shell; Git; LaTeX.
Familiar: Go; Java; R; SQL.

Coursework
Graduate: Comp Arch.; Adv. DB Systems; OS; Distributed Systems; Adv. Comp Arch; Adv. OS; Teaching CS Higher-Ed. **Undergraduate:** Algo Design and Analysis; Distributed Network Algos; Computer Networks; Computer Organization and Systems; OS; PLDI; Data Mining and Data Warehousing; DBMS; Hardware-based Memory Encryption (ISM); Computing in the Cloud.